LLM vs Manual vs EvoMaster

REST API Test Generation on Pre-seeded-Fault Benchmarks

Research Proposal · Pending instructor approval

Team SWT301_SU26_Group2 · Topic SE1916 · 2026-06-13 · v1.0

Dung (SE190034) · Huy (SE190240) · Dat (SE190239) · Thuan (SE190305) · Nguyen (SE190220)

The Problem & The Gap

LLMs can generate REST API tests from an OpenAPI spec — but is that output as good as a human's or EvoMaster's on faults we can actually count?

GAP-C

Comparison

No study puts LLM vs Manual vs EvoMaster on the same APIs. The only manual comparison (EvoMaster '19) is non-LLM — and scored below manual.

GAP-D

Dataset / Ground truth

Fault detection = raw 500-error counts on live APIs. No shared pre-seeded-fault set → ground-truth Recall is never computed.

GAP-M

Metric

Coverage-biased. No per-endpoint edge-case/error-code metric, no endpoint-type miss profile (CRUD / auth / error-handling).

Primary cluster: GAP-C + GAP-D · GAP-M = metric facet · (59 papers, 2018–2026)

Evidence — 59 papers (representative)

Paper	Tool / LLM	Metric	Best result
RESTGPT '24	GPT-3.5	rule P/R/F1	97% precision
KAT '24	GPT-3.5-turbo	status-code cov	74.9% (+15.7pp)
AutoRestTest '25	GPT-3.5 + MARL	code cov / 500s	42 500s vs EvoMaster 20
LlamaRestTest '25	Llama3-8B	code cov / faults	204 faults vs 130
RESTifAI '26	GPT-4.1-mini	operation coverage	128/134 ≈ 95.5%
EvoMaster '19	search-based	cov vs MANUAL	41% gen vs 82% manual

Pattern

- GPT dominates the LLM line.
- Metrics = coverage + 500-error counts only.
- Baselines are automated-only — no manual suite.
- Fault numbers are live-API counts — never Recall vs a known total.

Research Questions

Locked after approval — no changes (No-HARKing)

RQ 1 — Coverage

H1: LLM endpoint coverage > 90%

metric **endpoint coverage %**
threshold **≥ 90% (Case 2)**
test 1-sample Wilcoxon

RQ 2 — Fault detection

H1: Recall(LLM) > Manual & EvoMaster

metric **mutation-kill Recall**
threshold **comparative**
test Friedman + McNemar

RQ 3 — Edge cases

H1: edge-cases/endpoint LLM > Manual

metric **scenarios / endpoint**
threshold **comparative (Case 3)**
test paired Wilcoxon

Experiment Protocol

OpenAPI spec → GPT-4o few-shot → REST-assured tests → run vs original + mutants → coverage · Recall · edge-cases

Dataset	3 EMB APIs (rest-ncs, rest-scs, features-service) = 35 ops; mutation-seeded faults (ground truth)
LLM config	gpt-4o-2024-08-06 · temp 0 · few-shot from spec · prompt logged verbatim (pilot: Claude proxy)
Baselines	Manual (EP/BVA, blind) + EvoMaster 6.0.0 black-box (--maxTime 120s)
Stats	Wilcoxon / Friedman+McNemar / paired Wilcoxon · α=0.05 · Holm+Bonferroni · effect sizes

Timeline & Roles

Week 5–6
Proposal + GV approval

Week 7
Pilot done

Week 8
Full run (GPT-4o)

Roles (1 deliverable each · LR != MS enforced)

PL
Nguyen
Coordinate · merge · GV liaison

DG
Dat
Dataset · fault catalog · §3

LR
Dung
Harness · LLM/EvoMaster runs

MS
Huy
Metrics · statistical tests · §6

RW
Thuan
Threats · format · slides

Threats to Validity — top 3

Internal

LLM & Manual both Claude-authored in pilot (author bias)

→ Blind protocol (suites built before faults; spec-only sub-agent logs) + human cohort + GPT-4o for the full run

External

3 small EMB APIs; n=3 limits Friedman power

→ Per-fault McNemar pooled (N≈133) is the primary RQ2 test; EMB/JVM-only stated as a limit

Construct

Mutation kill ≠ real-fault detection; coverage ≠ quality

→ Standard PIT/Offutt operators (cited); report Recall + coverage + edge-cases jointly

Preliminary Pilot Signal (Week 7)

Feasibility evidence — full run with GPT-4o will confirm

100%

LLM endpoint coverage
(RQ1 · 35/35 ops · p≈1.6e-9)

217 vs 141

edge-case scenarios LLM vs Manual
(RQ3 · median 5 vs 4 · p≈6.2e-7)

0.135 vs 0.068

fault Recall EvoMaster vs LLM/Manual
(RQ2 · value-oracle catches logic faults)

Pilot used Claude Sonnet 4.6 as an available-model proxy; approved full run uses gpt-4o-2024-08-06.